

Houlin (Richard) Wang

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Computational linguist and speech scientist with 5+ years of experience engineering speech processing pipelines, evaluating foundational ASR models, and modeling non-standard speech variation. Proven track record in ASR benchmarking, forced alignment, low-resource NLP, and translating experimental findings into scalable speech AI technologies.

EDUCATION

University of California, Santa Cruz

Santa Cruz, CA

Doctor of Philosophy (Ph.D.) in Linguistics (Specialization: Speech Technology & Computation)

Expected June 2027

University of California, Los Angeles

Los Angeles, CA

Master of Arts (M.A.) in Linguistics (GPA: 3.97) | Bachelor of Arts (B.A.) in Linguistics, Magna Cum Laude (GPA: 3.95) *June 2022*

Specialization in Computation • Minor in Japanese • UCLA Department Scholar & Departmental Honors

RESEARCH & PROFESSIONAL EXPERIENCE

UC Santa Cruz

Santa Cruz, CA

Graduate Researcher — Speech Technology & Computational Linguistics

September 2022 – Present

- Engineer end-to-end acoustic feature extraction and automated signal processing pipelines in Python and R, accelerating corpus analysis throughput by 10x over manual baseline methods
- Model non-standard phonetic variation across 20,000+ mixed-effects regression iterations over 8 interacting structural variables, identifying key acoustic predictors to enhance ASR dialectal robustness
- Develop a language-agnostic detection framework for fine-grained speech reduction, establishing quantitative quality-monitoring metrics for speech training datasets and pronunciation lexicons
- Manage acoustic annotation, phone-level segmentation, and cross-speaker quality control across 30+ native speaker corpora, disseminating findings across 4 peer-reviewed publications

Living Tongues Institute for Endangered Languages

Remote

AI Speech Evaluation Research Fellow

June 2026 – August 2026

- Evaluated Meta's Omnilingual foundation ASR system across 5 low-resource language varieties, benchmarking WER/CER degradation between trained and zero-shot deployment domains
- Diagnosed systemic decoder failure modes, including hallucinations, infinite looping, script divergence, and training data mislabeling, delivering strategic deployment roadmaps to leadership
- Governed and annotated 90+ hours of speech audio data by establishing standardized quality-assurance guidelines and validation protocols across a distributed research team

Science Internship Program (SIP), UC Santa Cruz

Santa Cruz, CA

Research Mentor & Instructor (Computational Linguistics)

June 2026 – August 2026

- Designed and delivered an 8-week computational research curriculum, providing 80+ hours of structured training in Python/R, acoustic feature extraction, and quantitative hypothesis testing
- Supervised 3 student researchers in Praat automation and acoustic segmentation, guiding the annotation and mixed-effects regression modeling of 2,500+ speech tokens

UCLA Computational Phonology & Language Processing Labs

Los Angeles, CA

Computational Linguistics Research Assistant

September 2019 – June 2021

- Built Python algorithms to analyze phonotactic and morphophonological distributions across 10,000+ lexicon entries
- Constructed reproducible R pipelines for linguistic corpus analysis; coordinated experimental protocols and data ingestion for 30+ research participants

TECHNICAL SKILLS

Speech & Language Processing: Automatic Speech Recognition (ASR), Speech-to-Text (STT), ASR Evaluation (WER, CER), Forced Alignment (MFA, Praat), Acoustic Feature Extraction, Corpus Annotation, Dialect Modeling, LLMs, Low-Resource NLP

Machine Learning & Statistics: Mixed-Effects Regression (lme4), Hypothesis Testing, ML Classification, Bootstrapping, Model Comparison

Programming & Tools: Python (NumPy, SciPy, pandas), R (tidyverse, ggplot2), SQL, Praat, Git, Linux/Bash, LaTeX, PyTorch, Hugging Face

Languages: Mandarin Chinese (Native), English (Fluent), German (Working Proficiency), Japanese (Working Proficiency)

SELECTED PUBLICATIONS

Wang, R. (2025). The distribution of tone affects prosodic parsing in Beijing Mandarin: Evidence from retroflex lenition. *Proceedings of AMP 1*(1)

Wang, R. (2025). Morphosyntax-Prosody Mismatch in Beijing Mandarin: Evidence from Retroflex Lenition. *Proceedings of WCCFL 41*, 592–599

Wang, R. et al. (under review). Does Omnilingual work? Testing ASR functionality on training languages and zero-shot contexts. *ComputEL-10*

AWARDS & FELLOWSHIPS

THI Summer Public Fellowship (\$6,000, 2026) • **THI Summer Dissertation Fellowship** (\$5,000, 2026) • **SIP Fellowship** (\$4,750, 2026) • **Graduate Communications Fellowship** (\$1,400, 2023–2024) • **LSA Fellowship** (\$2,500, 2023)